

Python AI & ML - Answer Sheet

SECTION A – MCQ Answers

1. c) Guido van Rossum
2. b) Yes
3. d) None of the mentioned
4. a) Indentation
5. d) Parentheses, Exponential, Multiplication, Division, Addition, Subtraction
6. d) All of the mentioned
7. b) Unlimited length
8. c) [1, 2, 'hello']
9. b) 0
10. d) 56.24
11. c) The autonomous acquisition of knowledge through computer programs
12. c) They can learn complex details from data
13. a) Supervised learning requires labeled data
14. b) Naïve Bayes
15. b) True
16. b) Expectation Maximisation
17. b) Descriptive model
18. d) Both a and b
19. d) All of these
20. c) KNN has assumptions

SECTION B – Short Answers

- 1. Types of Machine Learning:** Supervised, Unsupervised, Reinforcement Learning.
- 2. Difference:** Supervised uses labeled data, Unsupervised finds patterns without labels.
- 3. Handling Missing Data:** Remove values, fill with mean/median, interpolation, ML prediction.
- 4. Choosing Classifier:** Small data → Naïve Bayes/KNN; Large data → Neural Networks.
- 5. ML Stages:** Data preprocessing, training, testing.
- 6. Deep Learning:** Neural networks with multiple layers for complex learning.
- 7. Classifier Selection:** Depends on dataset size and complexity.

SECTION C – Long Answers

- 1. Chess using RL:** AI learns by rewards (win/loss) through trial and error.
- 2. Amazon Recommendation:** Uses collaborative and content-based filtering.
- 3. Recommendation System:** Suggests items based on user behavior.
- 4. Training vs Test:** Train model vs evaluate model. Split: 70/15/15.
- 5. Applications:** Spam detection, medical diagnosis, fraud detection.